MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE 1SWF

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the cable described herein shall consist of this document and the latest issue of MIL-C-915.

REQUIREMENTS:

Qualification required.

Construction (watertight)

First - Copper, conductor, in accordance with ASTM B 286, size 22-7, uncoated. Maximum diameter 0.033 inch.
Second - Polyethylene insulation, natural, type II, class L, grade 3 in accordance with L-P-390. Nominal thickness 0.045 inch.
Third - Braided shield of AWG no. 36 uncoated copper. Braid angle of 30 to 35 degrees; minimum coverage of 85 percent.
Fourth - Shield insulation of one polyester tape, type G in accordance with MIL-I-631, plus a jacket of clear polyamide in accordance with ASTM D 4066. Nominal thickness of 0.003 inch. Standard identification code applied by method 2.
- Alternate shield insulation of two polyester tapes, type G in accordance with MIL-I-631, sealed. Standard identification code applied by method 2, on inner tape.
Fifth - Two conductors, one black and one white, cabled with a lay not greater than 3 inches. Nonfibrous fillers shall be used.
Sixth - Binder tape applied helically with overlap.
Seventh - Arctic tape polychloroprene jacket (black). Cable jacket thickness 0.100-inch maximum. Cable surface marking required.

TABLE I. Details.

Type and size	Number of conductors	Overall diameter	
		Minimum (inch)	Maximum (inch)
1SWF-2	2	0.600	0.625

FSC 6145

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EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum.	17.03
Voltage withstand - volts, root mean square, minimum	
Conductor to shield	3000
Shield to shield	500
Insulation resistance - megohms/1000 feet, minimum	
Conductor to shield	5000
Shield to shield	100
Group A:	
Visual and dimensional	No failure
Hydrostatic (open end) - leakage at 1000 lbf/in^2, in 6 hours, in^3, test each shipping length	Zero
Capacitance - at 1 kHz, pF/ft, maximum	28
Characteristic impedance - at 1 MHz, ohms	61 ± 3
Thermoplastic flow at 95 ± 3°C	No failure
Group B:	
Cold working (minus 54 ± 2°C)	No damage
Drip - at 75 ± 1°C	Zero
Physicals (unaged)	
Insulation (conductor)	
Tensile strength - lbf/in^2, minimum	1400
Elongation - percent, minimum	300
Jacket (cable)	
Tensile strength - lbf/in^2, minimum	1800
Elongation - percent, minimum	300
Set - inch, maximum	3/8
Group C:	
Physicals (aged)	
Insulation (conductor)	
Air oven	
Elongation - percent of unaged, minimum	60
Permanence of printing (jacket) - cycles, minimum	250
Cable filler removability	No failure
Shield-conformance to material coverage and construction	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

Cable aging and compatibility (95 ± 3°C)	No failure
Cold working (minus 54 ± 2°C)	No failure

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UNIT ORDERING LENGTHS:
All sizes 1000 feet (nominal).

Revision letters are not used to denote changes due to the extensiveness of the changes.

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Review activities:
Army - CR, ER
Navy - EC
DLA - IS

User activities:
Army - AL, AR, ME
Navy - CG

Preparing activity:
Navy - SH
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